

Bruno Paolo Barbaran Quiroz

Software Engineer | Full Stack & Mobile Developer

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PROFESSIONAL SUMMARY

Software Engineer with **3+ years of experience** building **Flutter mobile applications** and full stack web solutions for **infrastructure, municipal management, and fintech sectors**. Specialized in **Clean Architecture + BLoC**, complex system integrations, and process automation with AI and local OCR. Delivered end-to-end projects in production environments with a core stack of **Flutter, React, Python, and Node.js**.

PROFESSIONAL EXPERIENCE

CESEL Ingenieros

FullStack Programmer Analyst

November 2025 - Present

Lima, Peru | On-site

- Redesigned the **Portal Cesel** mobile app architecture (Flutter / Clean Architecture), implementing a **Quick Access system with PIN and biometric authentication** (fingerprint / Face ID) via `local_auth` and `flutter_secure_storage`, cutting authentication steps from **4 to 1** for returning users.
- Delivered the end-to-end **expense report module**: receipt capture with image upload, multi-step review workflow, and an **approval/rejection engine with email notifications**, covering **100%** of the corporate expense cycle.
- Migrated the document OCR pipeline from the external **OCR Space API** to a fully **self-hosted PaddleOCR solution**, eliminating **100% of third-party dependency** and reducing processing costs to **\$0/month**.
- Built a **SUNAT tax authority scraper** for automated supplier lookup, electronic invoice emitter verification, and **CPE document validation**, reducing tax compliance checks to **under 5 seconds** from a previously manual workflow.
- Architected and implemented the **Portal Cesel mobile backend** using a **Java (Spring Boot) microservices architecture**, separating domains (authentication, expense management, supplier services) and exposing REST APIs consumed directly by the Flutter mobile app.
- Provisioned **Linux servers** for OCR and scraping services and configured **CI/CD pipelines with GitHub Actions**, reducing production deployment time by **60%**.

Altoken.io

Software Architect

October 2025 - Present

Lima, Peru | Remote

- Designed the **end-to-end system architecture** of a **fintech / proptech platform** for real estate asset tokenization, defining service boundaries, integration contracts, and cross-cutting security standards.
- Implemented **Real-World Asset (RWA) tokenization logic** on **Hedera Token Service (HTS)**, programming token minting, transfer, and burn operations with full on-chain auditability via **Hedera Consensus Service (HCS)**.
- Integrated the **Hedera SDK** into the Node.js backend to manage accounts, sign transactions, query balances, and execute on-chain operations programmatically and securely.
- Built the platform's **REST APIs** (authentication, asset management, transactions, investor portfolio) with **JWT**, role-based access control, and AES encryption at rest and in transit.
- Developed the **digital wallet module** for investors: Hedera account provisioning, token portfolio display, and in-platform transaction signing.
- Architected the **KYC / onboarding flow** with identity validation, document verification, and status management to meet fintech regulatory compliance requirements.
- Built the **platform frontend** (React + TypeScript), including the investor dashboard, fractional asset purchase flows, and real-time tokenized portfolio visualization.

CECOM San Juan de Lurigancho

September 2024 - September 2025

- Led the full development lifecycle of **5+ municipal web applications** in production, from architecture to deployment, coordinating a 3-person engineering team.
- Built a **real-time tracking system** for Tetra radios using WebSockets, with an automatic zone-and-speed alert algorithm deployed for the district public safety operations center.
- Developed a **citizen identity validation platform** with real-time facial recognition, reducing municipal staff verification time to **under 3 seconds**.
- Designed and implemented an **attendance and HR management system** (React, TypeScript, Tailwind CSS), replacing fully manual processes and centralizing personnel administration city-wide.

Dominiotech

October 2022 - December 2023

Full Stack Web Developer

Lima, Peru

- Built an **interactive map dashboard** with Google Maps API displaying **200+ environmental monitoring stations**, with clustering and lazy loading to maintain load times **under 2 seconds** and dynamic filtering by type, status, and proximity.
- Developed an **internal NLP chatbot** (LangChain + Transformers) for conversational document search, reducing staff information retrieval time by **~70%**.
- Implemented a **bulk data import module** (CSV / Excel) for 200+ monitoring points with automatic validation, eliminating manual data entry and monthly reporting errors.
- Designed **real-time meteorological dashboards** integrating automated weather stations and forecasts, adopted as a planning tool for environmental field sampling and equipment maintenance.

EDUCATION

Universidad Tecnologica del Peru (UTP)

Graduated 2024

Professional Degree in Software Engineering | Lima, Peru

TECHNICAL SKILLS

Mobile: Flutter · Dart · Clean Architecture · BLoC/Cubit · local_auth · flutter_secure_storage · flutter_map · Android / iOS

Frontend: React · Next.js · Vite · TypeScript · JavaScript · Tailwind CSS · Material-UI · shadcn/ui

Backend: Java (Spring Boot) · Microservices · Node.js · Express.js · Python · Flask · .NET · REST APIs · JWT · Socket.io · Web Scraping

Blockchain & FinTech: Hedera Hashgraph (HTS/HCS) · RWA Tokenization · Digital Wallet · KYC/Onboarding · Smart Contracts

AI & Data: PaddleOCR · LangChain · Transformers · NLP · Machine Learning

Databases: PostgreSQL · SQL Server · MySQL · MongoDB · Firestore

DevOps: GitHub Actions · CI/CD · Docker · Linux Server · Git

Languages: Spanish (native) · English (upper-intermediate — technical reading & academic publication)

ADDITIONAL INFORMATION

Undergraduate Thesis: "Proposal for the Implementation of a Web Application with LLM for the Automation of Software Testing Execution and Incident Reports, Based on ISO/IEC/IEEE 29119-3" — Universidad Tecnologica del Peru | [UTP Repository](#)

Academic Publication: "Implementation of a Natural Language Processing Model for Spanish-Quechua Machine Translation" - Springer LNCS, 2024 | [10.1007/978-3-031-94770-4_37](#) | **Indexed in Scopus**

Community: Microsoft UTP